



Rolling of wheel without slipping

It states that any general displacement of a rigid body can be represented by a combination of translatory motion and rotational motion.

General plane motion = Translation + Rotation.

- 1) A road roller of mass 1200 kg and radius 0.25 m is pulled with a force $P = 1500 \text{ N}$ on a rough ground. If the roller starts from rest and rolls without slipping, determine the distance travelled by the centre of the roller C at which roller acquires a velocity of 3 m/s .

G. data:

$$m = 1200 \text{ kg}$$

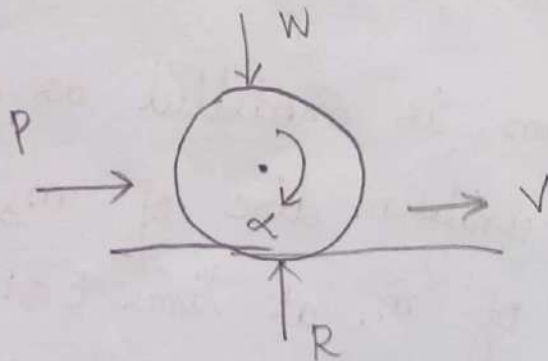
$$P = 1500 \text{ N}$$

$$r = 0.25 \text{ m}$$

$$v = 3 \text{ m/s}$$

$$u = 0 = \omega_0$$

$$s = ?$$



$$\sum F = P$$

$$\sum F = 1500$$

General plane motion = Translation + Rotation

$$\sum F \times s = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

$$I = \frac{1}{2} m r^2$$

$$= \frac{1}{2} \times 1200 \times 0.25^2$$

$$I = 37.5 \text{ kg} \cdot \text{m}^2$$

$$\omega = \frac{v}{r} = \frac{3}{0.25}$$

$$(\because v = 3 \text{ m/s})$$

$$\omega = 12 \text{ rad/s}$$

$$\Sigma F \times S = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

$$1500 \times S = \left(\frac{1}{2} \times 1200 \times 3^2 \right) + \left(\frac{1}{2} \times 37.5 \times 12^2 \right)$$

$$1500 S = 5400 + 2700$$

$$S = \frac{8100}{1500}$$

$$S = 5.4 \text{ m}$$

$\therefore$ The distance travelled by the roller is 5.4 m.

② An elevator system is modelled as $m_1 = 2000 \text{ kg}$, $m_2 = 1800 \text{ kg}$ and a uniform disc of $m_3 = 200 \text{ kg}$. Calculate the velocity of m_1 at time $t = 10 \text{ s}$ after the system starts from rest.

$$v_1 = v_2 = v$$

Q. data: $u = 0$.

$$m_1 = 2000 \text{ kg}$$

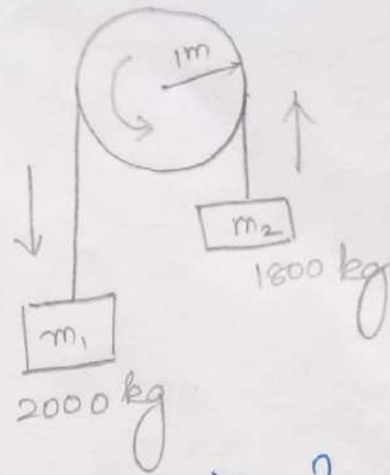
$$W_1 = 2000 \times 9.81 = 19620 \text{ N.}$$

$$m_2 = 1800 \text{ kg}$$

$$W_2 = 1800 \times 9.81 = 17658 \text{ N}$$

disk $m_3 = 200 \text{ kg}$

$$W_3 = 1962 \text{ N.}, \quad t = 10 \text{ s}, \quad v = ?$$



(3)

Applying Impulse-momentum Equation:

$$\Sigma F \times t = m(v - u).$$

Consider the block m_1 , (moving downward).

$$\Sigma F = W_1 - T_1 = 19620 - T_1$$

$$(19620 - T_1) 10 = 2000(v_1 - 0).$$

$$19620 - T_1 = 200v_1 \quad \text{--- (1)}.$$

Consider the block m_2 , (moving upward).

$$\Sigma F = T_2 - W_2 = T_2 - 17658$$

$$(T_2 - 17658) 10 = 1800(v_2 - 0).$$

$$T_2 - 17658 = 180v_2 \quad \text{--- (2)}.$$

Disc :

(4)

$$\sum M_G \times t = I_G (\omega - \omega_0)$$

$$I_G = \frac{mr^2}{2} = \frac{200 \times 1^2}{2}$$

$$= 100$$

$$\omega_0 = 0$$

($\because r = 1 \text{ m}$)

$$\begin{aligned} \sum M_G &= T_1 r - T_2 r = (T_1 - T_2) r \\ &= (T_1 - T_2) 1 \end{aligned}$$

$$(T_1 - T_2) 1 \times 10 = 100 (\omega - 0)$$

$$\omega = \frac{v}{r} = \frac{v}{1}$$

$$\therefore \omega = v$$

$$(T_1 - T_2) = \frac{100}{10} (v)$$

$$T_1 - T_2 = 10v \quad \text{--- (3)}$$

$$\therefore v_1 = v_2 = v$$

Add Eqn (1) & (2);

$$19620 - T_1 + T_2 - 17658 = 200v_1 + 180v_2$$

$$1962 - T_1 + T_2 = 380v$$

$$-T_1 + T_2 = 380v - 1962 \quad \text{--- (4)}$$

$$\text{Add (3) + (4)} \quad -T_1 + T_2 = -1962 + 380v$$

$$T_1 - T_2 = 0 + 10v$$

$$\hline 0 = -1962 + 390v$$



$$390 \text{ V} = 1962$$

(5)

$$V = \frac{1962}{390}$$

$$V = 5.03 \text{ m/s}$$

$$\text{Since } V_1 = V_2 = V,$$

Hence, the velocity of the mass $m_1 = 5.03 \text{ m/s}$.

Unit - 4 (University Q. paper prob.)

- 1) Two trains A and B leave the same station on parallel lines. A starts with a uniform acceleration of $\frac{1}{6} \text{ m/s}^2$ and attains a speed of 24 km/hr , when steam is reduced to keep the speed constant. B leaves 40 sec. after, with uniform acceleration of $\frac{1}{3} \text{ m/s}^2$ to attain a maximum speed of 48 km/hr . When will it overtake A?

Consider the motion of train A,

$$u = 0$$

$$a = \frac{1}{6} \text{ m/s}^2$$

$$V = 24 \text{ km/hr} = \frac{24 \times 1000}{3600} = 6.67 \text{ m/s}$$

$$V = u + at$$

$$6.67 = 0 + \frac{1}{6}t$$

(b)

$$t = 6.67 \times 6 = 40 \text{ s.}$$

Distance travelled by the train A in 40 s,

$$S = ut + \frac{1}{2}at^2$$

$$S = 0 + \left(\frac{1}{2} \times \frac{1}{6} \times 40^2 \right)$$

$$S = 133.33 \text{ m.}$$

∴ Distance travelled by the train A after 40 s,

↓

$$V = \frac{S}{t}$$

$$t = t - 40$$

$$S = 6.67(t - 40)$$

∴ Total distance travelled by A in t sec,

$$S = 133.33 + 6.67(t - 40)$$

$$= 133.33 + 6.67t - 266.8$$

$$S = 6.67t - 133.47 \quad \text{--- (1)}$$

Consider the motion of train B [train B starts after 40 s of train A)

$$u = 0$$

$$a = \frac{1}{3} \text{ m/s}^2$$

$$V = 48 \text{ km/hr.}$$

$$= \frac{48 \times 1000}{3600} = 13.34 \text{ m/s.}$$

$$V = u + at$$

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$$13.34 = 0 + \frac{1}{3} t$$

$$t = 40 \text{ s}$$

As train B starts after 40 s,

$$t = t + 40$$

$$= 40 + 40 = 80 \text{ s}$$

Distance travelled by the train B in 40 s,

$$s = ut + \frac{1}{2} at^2$$

$$= 0 + \left(\frac{1}{2} \times \frac{1}{3} \times 40^2 \right)$$

$$= 266.67 \text{ m}$$

But, the time taken by train B is 80 sec, hence, the distance travelled by the train B after 80 s,

$$V = \frac{s}{t}$$

↓
t-80

$$13.34 = \frac{s}{t-80}$$

$$s = 13.34(t-80)$$

∴ Total distance travelled by B in t sec,

$$s = 266.67 + 13.34(t-80)$$

$$= 266.67 + 13.34t - 1067.2$$

$$s = 13.34t - 800.53$$

At the instant, when the train B overtakes the train A, distance travelled by both the trains will be equal. Hence, Equate (i) and (ii).

$$6.67t - 133.47 = 13.34t - 800.53$$

$$6.67t - 13.34t = -800.53 + 133.47$$

$$\cancel{6.67t} = \cancel{6.67} \cdot 06$$

$$t = \frac{667.06}{6.67}$$

$$\underline{\underline{t = 100 \text{ sec}}}$$

(2) Two stations P and Q are 5.2 km apart. An automobile starts from rest from the station 'P' and accelerates uniformly to attain a speed of 48 kmph in 30 sec. This speed is maintained until the brakes are applied. The automobile comes to rest at the station Q with an uniform retardation of one meter per sec. Determine the total time required to cover the distance between these two blocks.

$$u = 0, \quad v = 48 \text{ km/hr.}$$

$$= \frac{48 \times 1000}{3600} = 13.33 \text{ m/s.}$$

$$S = 5.2 \text{ km} = 5.2 \times 1000 \\ = 5200 \text{ m.}$$

$$t = ? = t_1 + t_2 + t_3 \\ S_1 = ? , S_2 = ? , S_3 = ?$$

$$\text{acc.} = \frac{\text{velocity}}{\text{time}}$$

$$\tan \theta = \frac{\text{opp.}}{\text{adj.}}$$

$$\text{since } a = 1 \text{ m/s}^2 \therefore \tan \theta = 1 \text{ m/s}^2.$$

When brakes are applied the automobile retards from a velocity of 13.33 m/s to zero in $t_3 \text{ sec}$ at the rate of 1 m/s^2 . Then,

$$\tan \theta = \frac{13.33}{t_3}$$

$$t_3 \tan \theta = 13.33$$

$$t_3 \times 1 = 13.33 \Rightarrow t_3 = 13.33 \text{ s.}$$

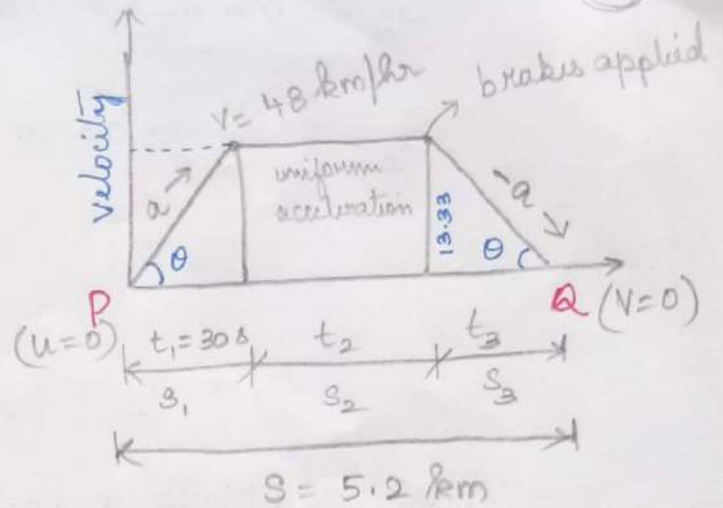
Let t_2 be the time during which the automobile travels at uniform velocity,

$$S = S_1 + S_2 + S_3$$

$$5200 = S_1 + S_2 + S_3$$

$$S_1 = \frac{1}{2} \times b \times h = \frac{1}{2} \times 30 \times 13.33$$

$$= 199.95 \text{ m.}$$



$$S_2 = b \times h = 13.33 \times t_2$$

(10)

$$S_3 = \frac{1}{2} \times b \times h = \frac{1}{2} \times 13.33 \times 13.33 \\ = 88.84$$

$$\therefore S = S_1 + S_2 + S_3$$

$$5200 = 199.95 + 13.33 t_2 + 88.84$$

$$13.33 t_2 = 5200 - 199.95 - 88.84$$

$$13.33 t_2 = 4911.21$$

$$t_2 = \frac{4911.21}{13.33}$$

$$t_2 = 368.43 \text{ s.}$$

$\therefore$ total time required to cover the distance between the two stations. ($t = t_1 + t_2 + t_3$).

$$t = 30 + 368.43 + 13.33$$

$$t = 411.76 \text{ sec.}$$